MATH 142: Elementary Algebraic Topology

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11/6/26 Lecture 30: Orientations of Simplices

Let X be a topological space. We want invariants of X which are homotopy invariant. We have π_0 and π_1. Using π_0, we showed $\mathbb{R} \not\cong \mathbb{R}^n$ for $n > 1$, and using π_1, we showed $\mathbb{R}^2 \not\cong \mathbb{R}^n$ for $n > 2$.

One way to extend to $\mathbb{R}^k$ is using the *higher homotopy groups* $\pi_k(X, x)$ for $k \geq 0$, given by

$$\pi_k(X, x) = \{\text{homotopy classes of maps } S^k \to X \text{ sending the north pole to } x\}$$

Fact: This has a group structure, and for $k > 1$, these are abelian.

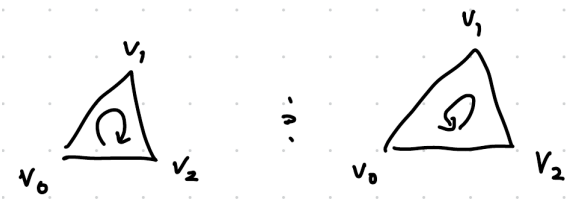
Problem: Hard to compute. We don't even know $\pi_k(S^2, x)$ for all k.

Our Approach: Simplicial Homology.

Pro: Relatively easy to compute.

Orientation of a Simplex

We want to distinguish



First case: Denote it with info (v_0, v_1, v_2) or a cyclic permutation of it.

Second case: Denote it with info (v_0, v_2, v_1) or a cyclic permutation of it.

Definition 30.1: Orientation

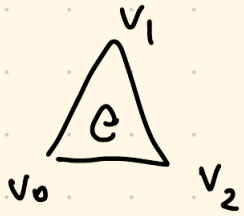
Let A be a simplex other than a point, and consider orderings of its vertices. Say two orderings are *equivalent* if they differ by an even permutation. The two equivalence classes are called *orientations* of A.

Definition 30.2: Induced Orientation

Say the vertices of A have an orientation given by $(v_0, v_1, \ldots, v_k)$. Say B is the face obtained by deleting v_i. We get an ordering $(v_0, \ldots, \widehat{v_i}, \ldots, v_k)$ of the vertices of B (the notation means remove v_i).

If i is even, say this is the *induced orientation* of B. If i is odd, it's the opposite of the induced orientation.

Example: A Triangle



The orientation of A is (v_0, v_1, v_2). Then, (v_1, v_2) is right, (v_0, v_2) is wrong so (v_2, v_0) is right, and (v_0, v_1) is right:

